AUREX · BCCL'S SECURE AI WORKSPACE

Reading data

A working method for getting from a spreadsheet you did not build to a number you would defend in front of your MD.

DISHANT SINGLA

B.Tech, IIT Roorkee 2015 · Founder, Base42 Systems

₹110 Cr	1 Bn+	200 M	120 M
ad business on the platform I built	ad impressions a month	monthly active users	Ola customer profiles built

Tuesday 4 August 2026 · 2:30–4:30 PM IST · Live, hands-on · Bring a laptop

WHO IS RUNNING THIS SESSION

I have spent eight years building the systems these numbers come out of.

Ola, Koo, Khatabook, Wealthy, HT Media. Customer data platforms, identity resolution, recommendation systems, ad targeting. Most recently the central data platform at HT Digital Streams, sitting under a ₹110 crore a year ad business.

Every trap in this session is one I have shipped, been burned by, or caught in review. That is the only reason it is in here.

AT SCALE

₹110 Cr
ad revenue running on the platform

1 Bn+
ad impressions a month

200 M
monthly active users

120 M
Ola customer profiles built

₹2.5 Cr
a year in retired tooling

60%
duplicate IDs eliminated

Why a company pays for this

The gap between a number that survives scrutiny and one that does not

The number nobody checked

A board pack goes out with a metric computed on the wrong date column. Three months of decisions get anchored to it.

The definition nobody agreed

Two teams present revenue in the same meeting, ₹40 Cr apart. Both are right. Nobody knows that yet.

The trend that was three accounts

A segment is declared a growth engine. One client churns and the growth engine goes with it.

WHAT CHANGES AFTER TODAY

Open a file you have never seen. Work out what it actually contains. Ask it eight questions that surface something. Then put a number in front of a decision-maker with the caveat already attached.

None of that needs Excel skill or statistics training. It needs the sequence, and it needs to know where numbers go wrong.

Aurex does the execution.

You do the judgement.

Five frameworks, one file, and the cards to run them again

Everything in this session is designed to be reusable on Monday

01

THE SEVEN-STEP READ

The sequence for any unfamiliar file

02

THE EIGHT LENSES

Every question a dataset can answer

03

THE CHART SELECTOR

Say the comparison, get the chart

04

THE SIX DISTORTIONS

Where confident numbers go wrong

05

THE FIVE-CHECK GATE

What to verify before you send

YOU ALSO LEAVE WITH

The practice file

3,079 rows of real-shaped ad bookings, traps included

Eight prompts

One per lens, written to force checkable output

The gate card

Five checks, one page, stick it on your monitor

One file. Every framework runs on it.

ad-sales-data.csv · three years of advertising bookings, 2022 to 2024

3,079	14	8	2,490	₹44.8 Cr
rows	columns	cities	confirmed bookings	net revenue

WHAT IS IN IT

Dates	booked and published	**Cities**	8 Indian metros
Editions	4 publications	**Ad types**	Display, Digital, Classified, and two more
Categories	10 advertiser sectors	**Reps**	30 salespeople
Money	rate, discount, gross, net	**Status**	confirmed, cancelled, pending

It carries four planted data-quality faults. Finding them is Step 2.

MODULE 1 OF 5

01

The Seven-Step Read

Analysts do not start with the numbers. They start with the decision, then the shape of the file, and only then the arithmetic.

STEPS 1 TO 7 · 25 MINUTES

Seven steps, in this order

WIDE to NARROW

1	**CONTEXT**	Whose decision, and when do they need it?
2	**SHAPE**	Rows, columns, date range, what is broken.
3	**TOTAL**	Start at the biggest number.
4	**BREAK DOWN**	One dimension at a time.
5	**COMPARE**	Against last year, the plan, or a peer.
6	**DRIVERS**	Why it moved. The parts have to add up.
7	**VALIDATE**	Would I put my name on this?

Step 1 — CONTEXT: three questions before the file opens

Most weak analysis is a correct answer to a question nobody had.

1. **Who decides, and when?**
   If you cannot name the person and the date, you are producing trivia.

2. **Which answer would change the decision?**
   If every possible finding leads to the same action, do not run the analysis.

3. **What is the smallest version you could answer today?**
   One metric, one dimension, one period. Widen later if it earns it.

What Step 1 sounds like in practice

Same file, same tool, two different opening moves

WITHOUT A FRAME

“Analyse this file.”

Revenue rose 4.3%

8 cities represented

3,079 rows

Q3 shows a dip

Average deal ₹1.8 lakh

All of it true. None of it usable.

WITH A FRAME

“Which two cities should lose headcount next quarter?”

One table

Revenue per rep, by city

Two cities named

A caveat about deal mix

One question, and the file has a job.

What Step 2 turned up in 3,079 rows

13 spellings of 8 city names

"Mumbai", "mumbai", "MUMBAI", "Mumabi", " Delhi", "Delhi "

10 rows with blank revenue

Every SUM you run drops them without saying so

5 bookings at 95% discount

A typing error, or a deal somebody needs to explain

589 non-confirmed rows

Cancelled and Pending. Leave them in and every total is inflated.

WAR STORY · HT DIGITAL STREAMS · IDENTITY RESOLUTION

We were counting the same person up to four times.

Three sites and a set of APIs, each minting its own user ID. Every audience number the business planned against was inflated, and nobody could see it, because each system was internally consistent.

The fix was not analytical. It was going back to what one row meant, across four systems that had each quietly decided for themselves.

60%

of duplicate IDs eliminated once the grain was defined once, centrally

Step 2 exists because “what is one row?” is the question that was never asked.

Spend the first two minutes on what the file is.

Somebody filtered it, named the columns and decided what to leave out before it reached you. Those decisions are now your assumptions, whether you look at them or not.

MODULE 2 OF 5

02

The Eight Lenses

Almost every question you will be handed is one of these eight.

Naming the lens tells you the cut, the chart, and the trap to check.

LENSES 1 TO 8 · 40 MINUTES

The Eight Lenses, on one page

The reference artefact for this module. Photograph this slide.

LENS	THE QUESTION	THE CUT	THE TRAP IT HIDES
1 **Comparison**	How does this stack up against that?	Total by dimension, sorted	Counts where rates belong
2 **Ranking**	Who is on top, who is at the bottom?	Sort both ends, not just the top	A top-10 list with no bottom
3 **Trend**	What changed, and when exactly?	Index to a common base	Two scales on one chart
4 **Composition**	What makes up the whole?	Share of total, by period	Mix shift read as performance
5 **Distribution**	What does a typical one look like?	Histogram, plus the median	The mean quoted as typical
6 **Correlation**	Do these move together?	Scatter, then a coefficient	Correlation stated as cause
7 **Outliers**	What looks wrong?	Flag against the median	Concentration read as a trend
8 **Decomposition**	Why did the number move?	Bridge that sums to the total	A residual nobody names

1 COMPARISON

THE QUESTION IT ANSWERS

How does this one stack up against that one?

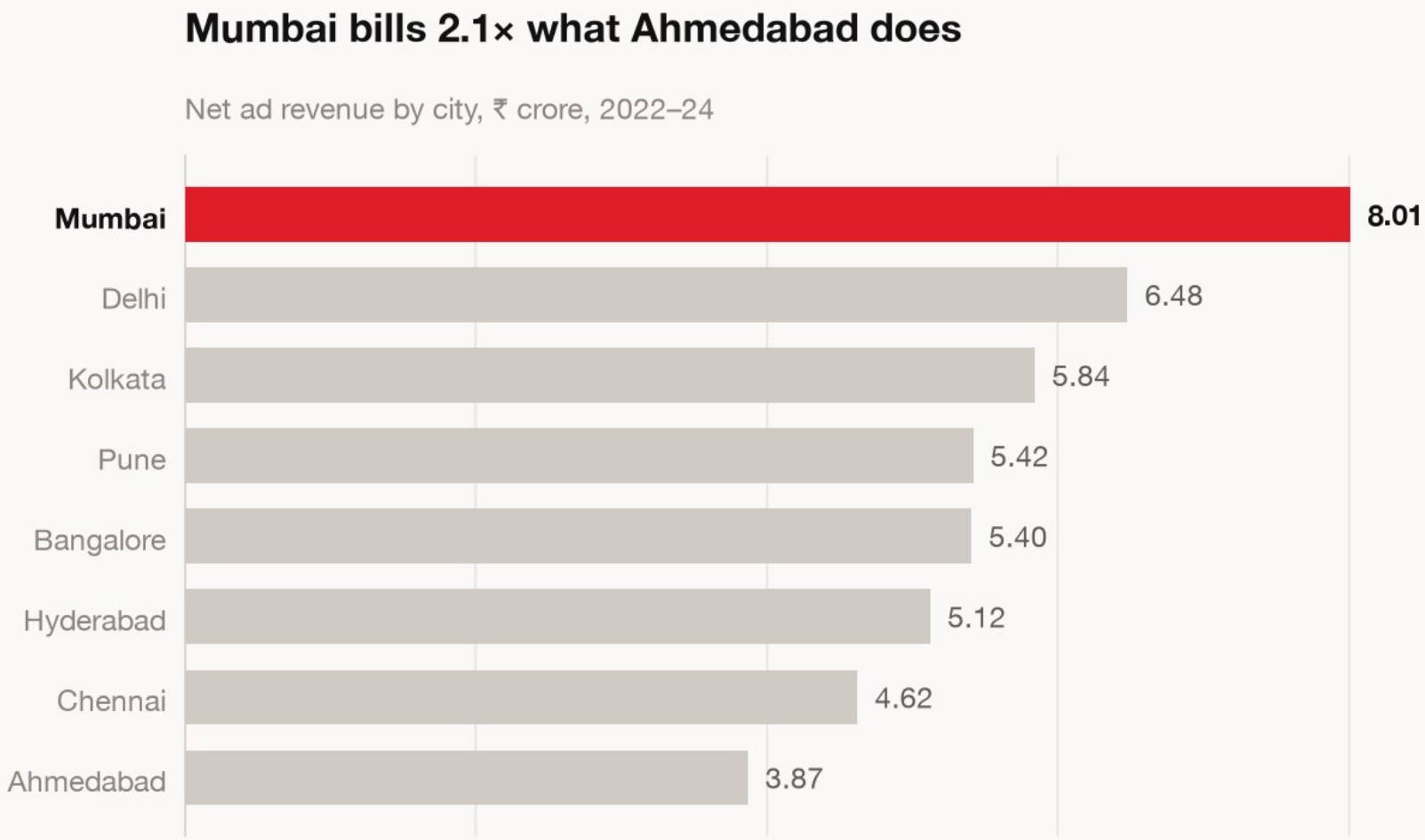
ON TODAY'S FILE

Which city bills the most ad revenue?

THE TRAP IT HIDES

Mumbai is biggest at everything. Biggest is not the same as best.

Next: the same lens run on 2,490 confirmed bookings.



Sorted, one colour on the point, numbers on the bars. You will reach for this shape more than any other.

The prompt behind Lens 1

Paste this, do not type it. The wording is the lesson.

```
Using the code interpreter on ad-sales-data.csv:

Filter to status = Confirmed and drop rows
where net_amount is blank. Tell me how many
rows you dropped.

The city column has spelling variants. Show me
the mapping you intend to apply before you
apply it.

Then total net_amount by city, sorted high
to low. Give me the gap between top and
bottom as a multiple, not just two numbers.

Return a horizontal bar chart, sorted, one
accent colour, values on the bars, titled with
the finding as a sentence.
```

WHAT COMES BACK

- A sorted bar chart, titled with the finding
- A cleaning log: 13 city strings collapsed to 8
- The row count it dropped, stated up front
- The gap as “2.1×”, not as two raw numbers

2 RANKING

THE QUESTION IT ANSWERS

Who is on top, and who is at the bottom?

ON TODAY'S FILE

Which reps carry the business, and which coast?

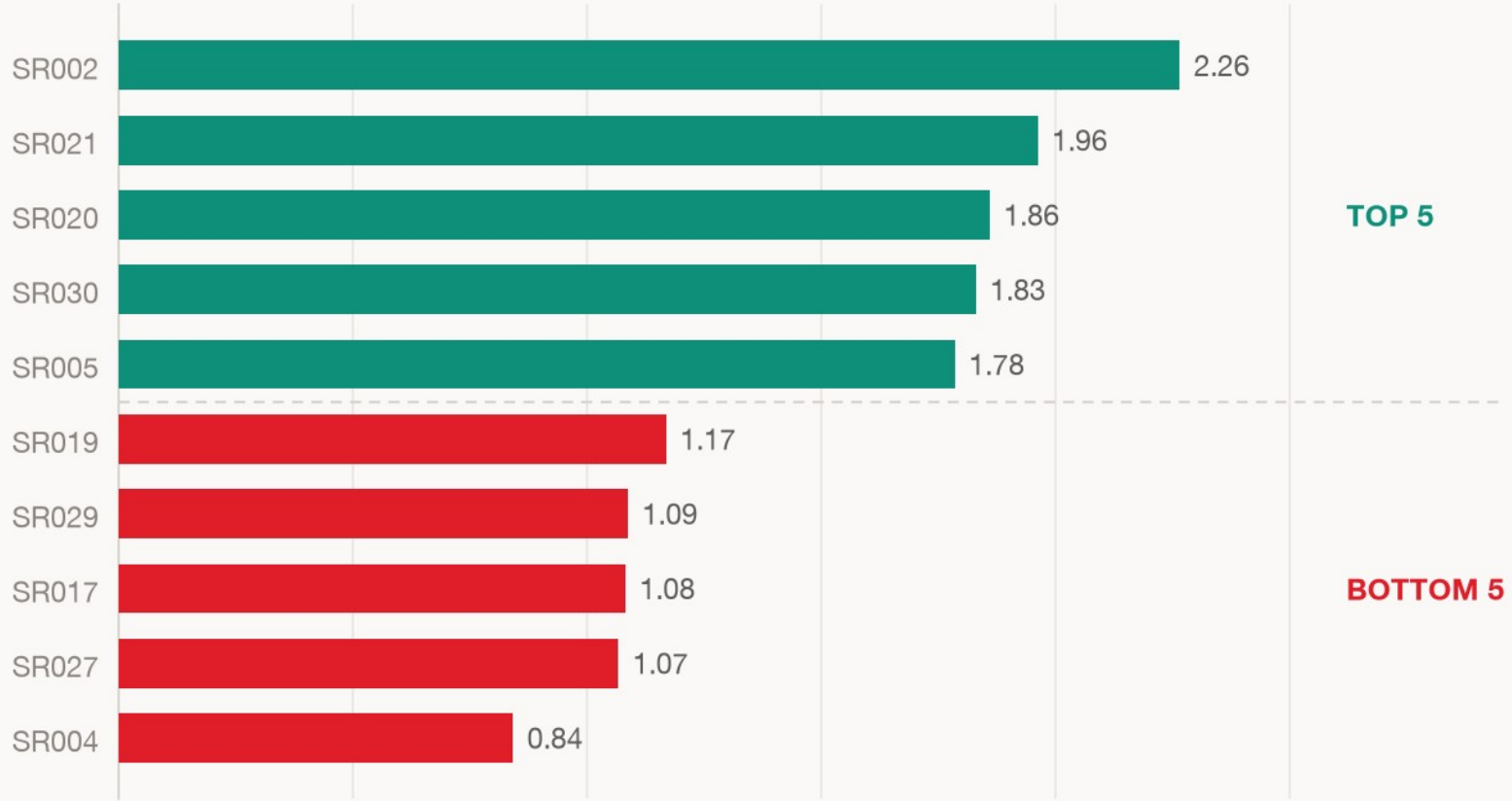
THE TRAP IT HIDES

A top-10 list flatters. The bottom is where the recoverable money sits.

Next: the same lens run on 2,490 confirmed bookings.

Your best rep bills 2.7× your weakest

Top 5 and bottom 5 of 30 reps · ₹ crore, 3-year total



Show both ends. A top-5 on its own hides the gap you can actually do something about.

3 TREND

THE QUESTION IT ANSWERS

What changed over time, and when exactly?

ON TODAY'S FILE

Is digital really growing, or does it just feel that way?

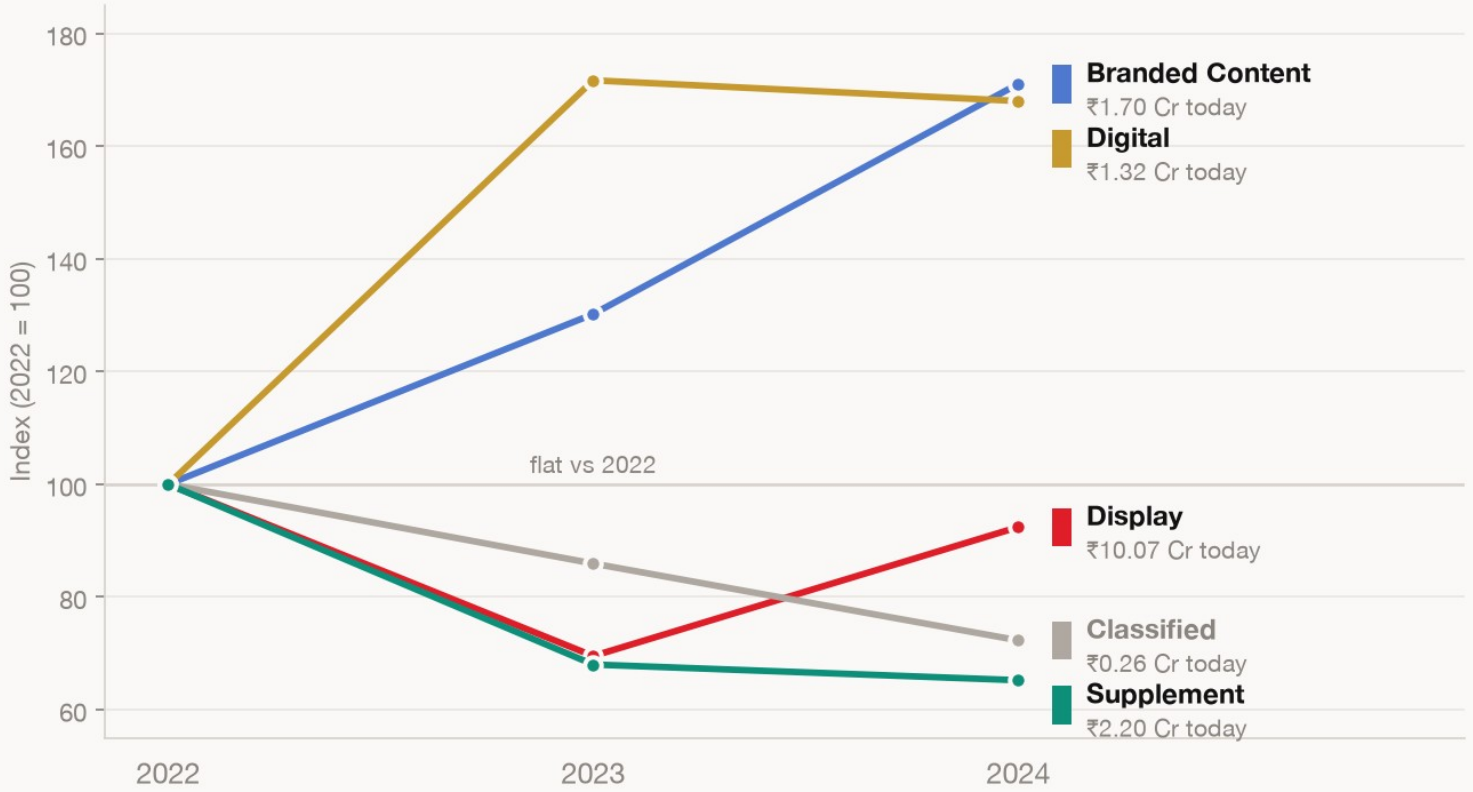
THE TRAP IT HIDES

Raw lines let the biggest series bury every other story.

Next: the same lens run on 2,490 confirmed bookings.

Branded Content and Digital are up ~70%. Print supplements are down 35%.

Net revenue indexed to 2022 = 100, with today's absolute size on the right



Indexed to 2022 = 100. When one series is 40× another, plotting raw values shows you the big line and nothing else.

4 COMPOSITION

THE QUESTION IT ANSWERS

What makes up the total, and is the mix moving?

ON TODAY'S FILE

Print is shrinking. By how much, and into what?

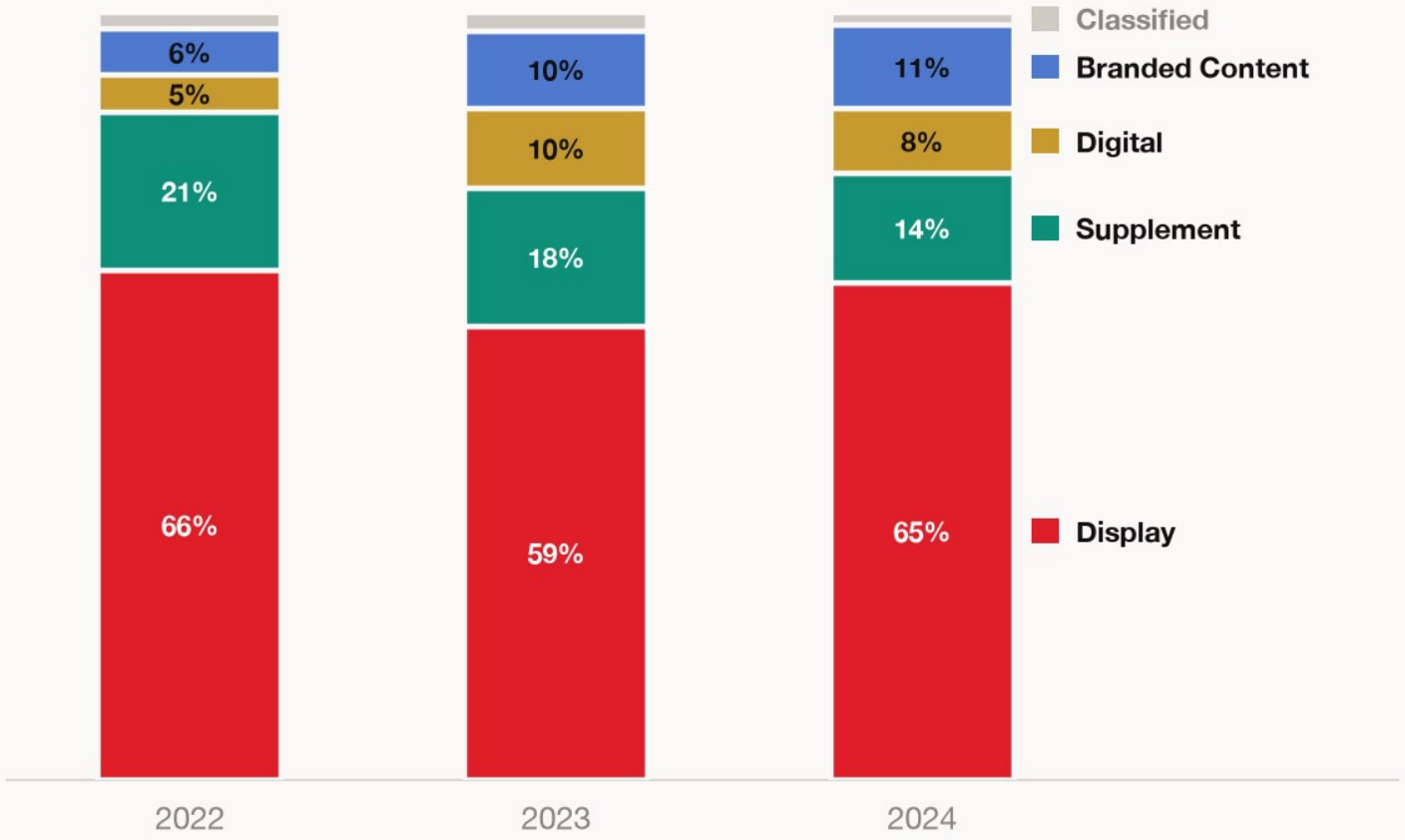
THE TRAP IT HIDES

Volume can rise while revenue stalls. That is mix, not performance.

Next: the same lens run on 2,490 confirmed bookings.

Digital + Branded Content grew from 12% to 20% of the mix

Share of net revenue by ad type · % of year total



Mix shift lives here. More of what sold was the cheap thing, so pages went up and money did not.

5 DISTRIBUTION

THE QUESTION IT ANSWERS

What does a typical one actually look like?

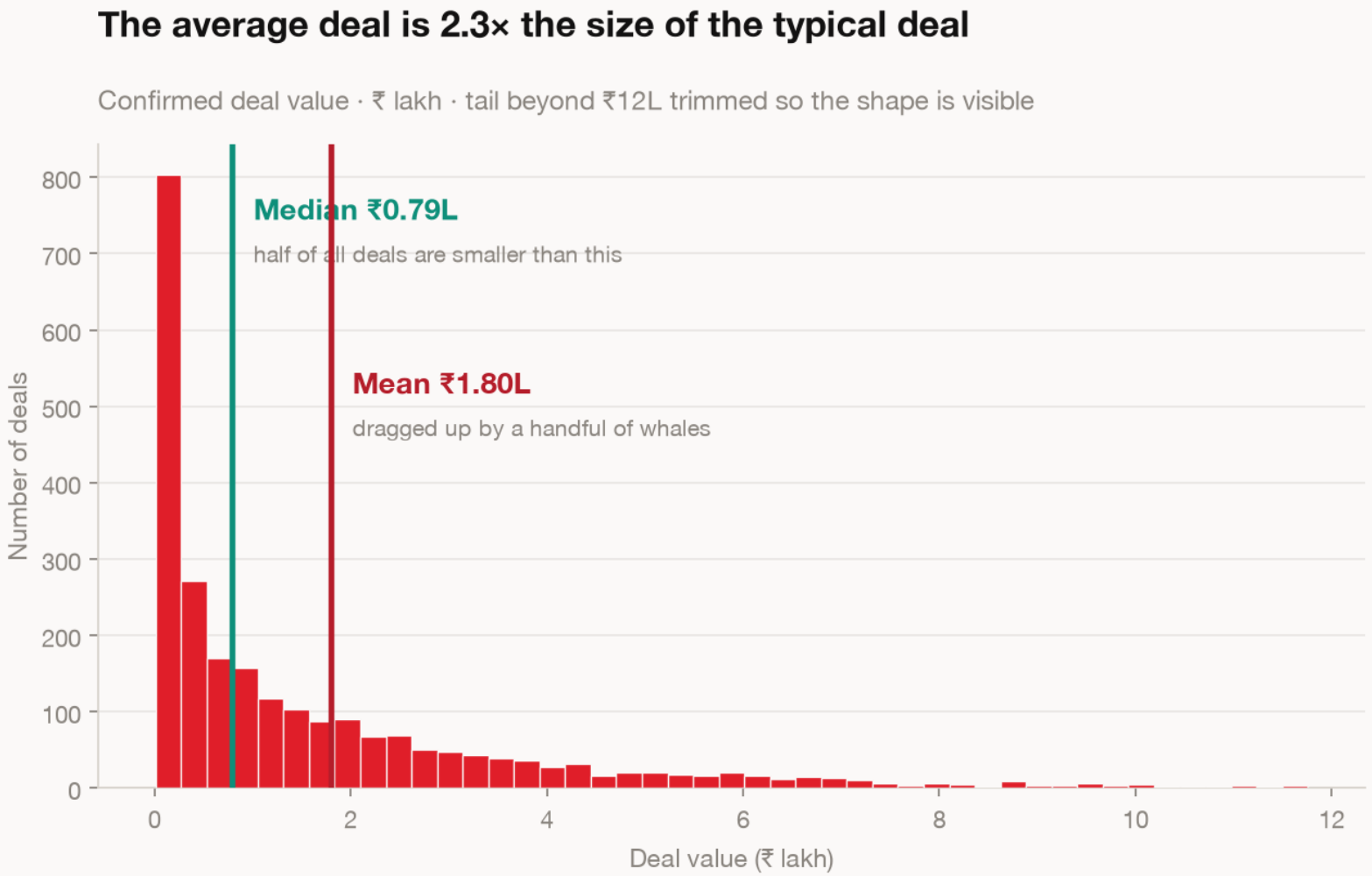
ON TODAY'S FILE

What is a normal deal size here?

THE TRAP IT HIDES

A handful of whales drag the average away from reality.

Next: the same lens run on 2,490 confirmed bookings.



Mean ₹1.80L, median ₹0.79L. Call the mean “typical” in a meeting and you are overstating by 2.3×.

6 CORRELATION

THE QUESTION IT ANSWERS

Do these two things move together?

ON TODAY'S FILE

Do bigger discounts actually win bigger deals?

THE TRAP IT HIDES

Two things moving together is not one causing the other.

Next: the same lens run on 2,490 confirmed bookings.



r = −0.05, so there is nothing here. That is still a result, and it is the one that settles a discount argument.

7 OUTLIERS

THE QUESTION IT ANSWERS

What looks wrong, and is it real or a data error?

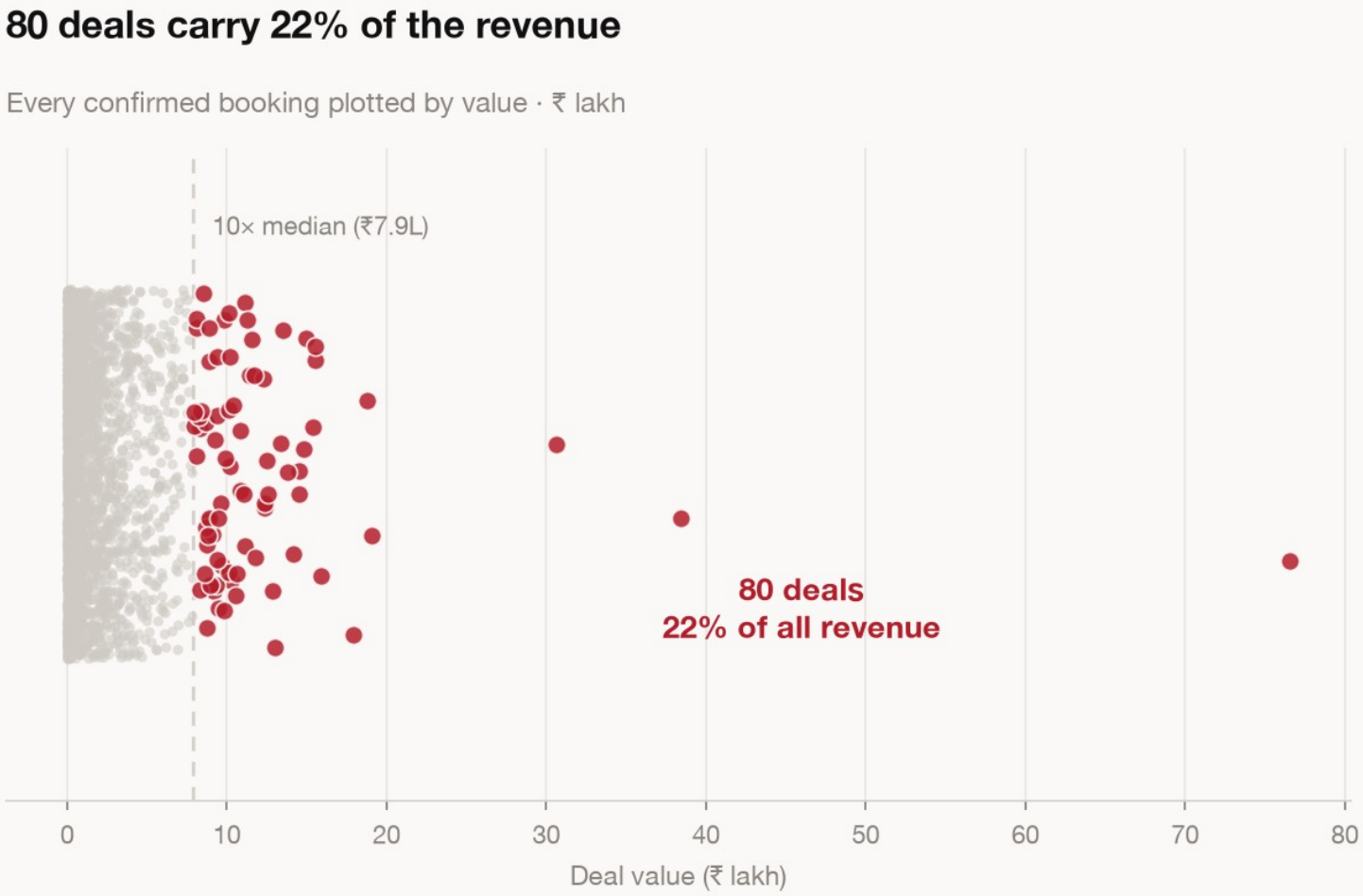
ON TODAY'S FILE

Are a handful of deals carrying the whole year?

THE TRAP IT HIDES

A few large deals can manufacture a trend that does not exist.

Next: the same lens run on 2,490 confirmed bookings.



80 deals out of 2,490 carry 22% of revenue. Lose two of them and the trend you were describing goes away.

8 DECOMPOSITION

THE QUESTION IT ANSWERS

Why did the number move?

ON TODAY'S FILE

Revenue grew 21%. More deals, or bigger ones?

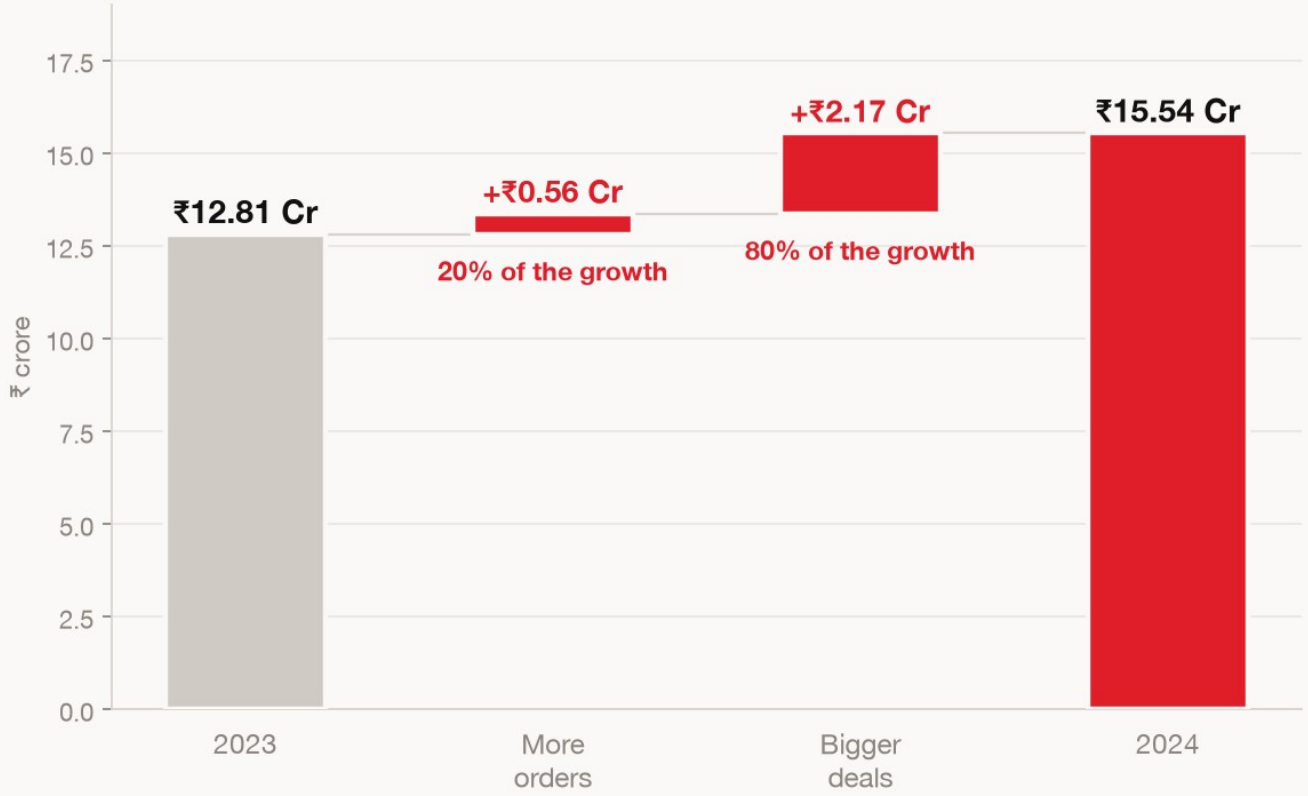
THE TRAP IT HIDES

"It went up" is not a driver. Parts that do not sum are not an answer.

Next: the same lens run on 2,490 confirmed bookings.

Revenue grew 21%. Bigger deals did four-fifths of it.

Net revenue bridge, 2023 to 2024 · ₹ crore · the parts sum to the total



The parts have to sum to the whole. An unexplained residual means you do not understand the movement yet.

The prompt behind Lens 8

Paste this, do not type it. The wording is the lesson.

```
Net revenue moved from ₹12.81 Cr in 2023 to
₹15.54 Cr in 2024. Find out why, with code.

1  Confirm the total change, absolute and %.

2  Split it into two effects:
     volume    = (orders24 − orders23) × avg23
     deal size = (avg24 − avg23) × orders24

3  The two must sum to the total change.
   Print the reconciliation and the residual.
   If the residual is not about zero, stop and
   tell me what is unexplained.

4  Label every driver OBSERVED or INFERRED.
```

WHAT COMES BACK

- A bridge where the parts tie to the total
- 20% from more orders, 80% from bigger ones
- The residual, printed, at about zero
- An honest label on anything it assumed

WAR STORY · HT DIGITAL STREAMS · CONTEXTUAL TARGETING

The lift was real. The first number we ran was not.

We moved from broad targeting to in-page semantics with brand-safety controls. The early read looked enormous, because the two groups were not comparable. Different placements, different weeks, different mix.

Once it was cut properly against a like-for-like baseline, the gain held up. Smaller than the first number, and defensible in front of a client.

+25%

CTR and conversion against vanilla targeting, on a like-for-like cut

Lens 1 is the whole discipline: a number means nothing until you name what it is compared to.

Most people stop at “revenue went up.”

Lenses 6 to 8 are the ones that get skipped, and the ones that get you invited back into the room.

HANDS-ON · 12 MIN

- Open ad-sales-data.csv in Aurex.
- Pick any two lenses and run them properly.
- Write down which trap you checked for.

Post in chat: “The one thing that surprised you most”

MODULE 3 OF 5

03

The Chart Selector

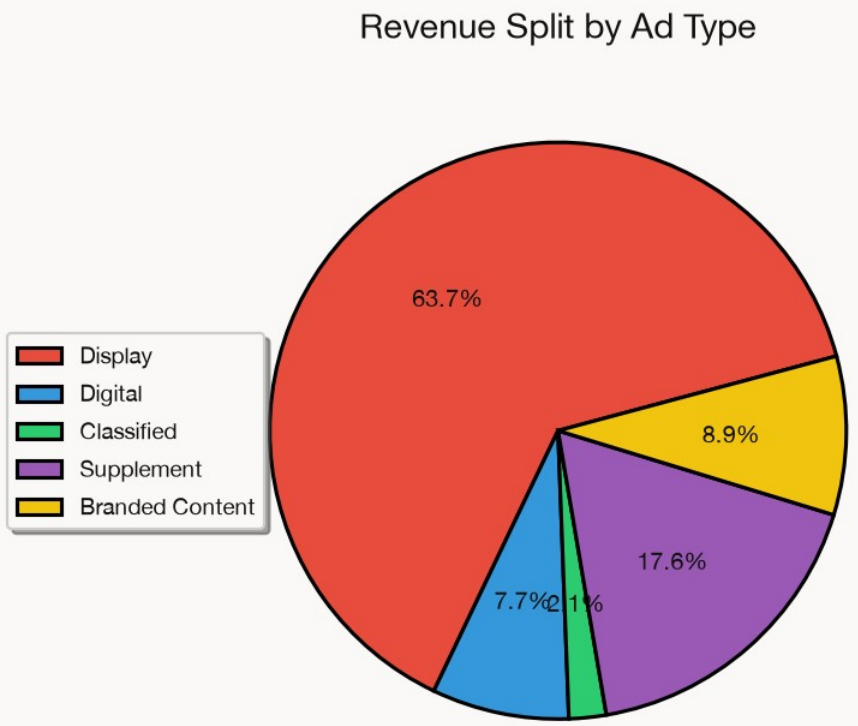
Nine comparisons, nine answers. Name the comparison out loud and the chart stops being a matter of taste.

15 MINUTES

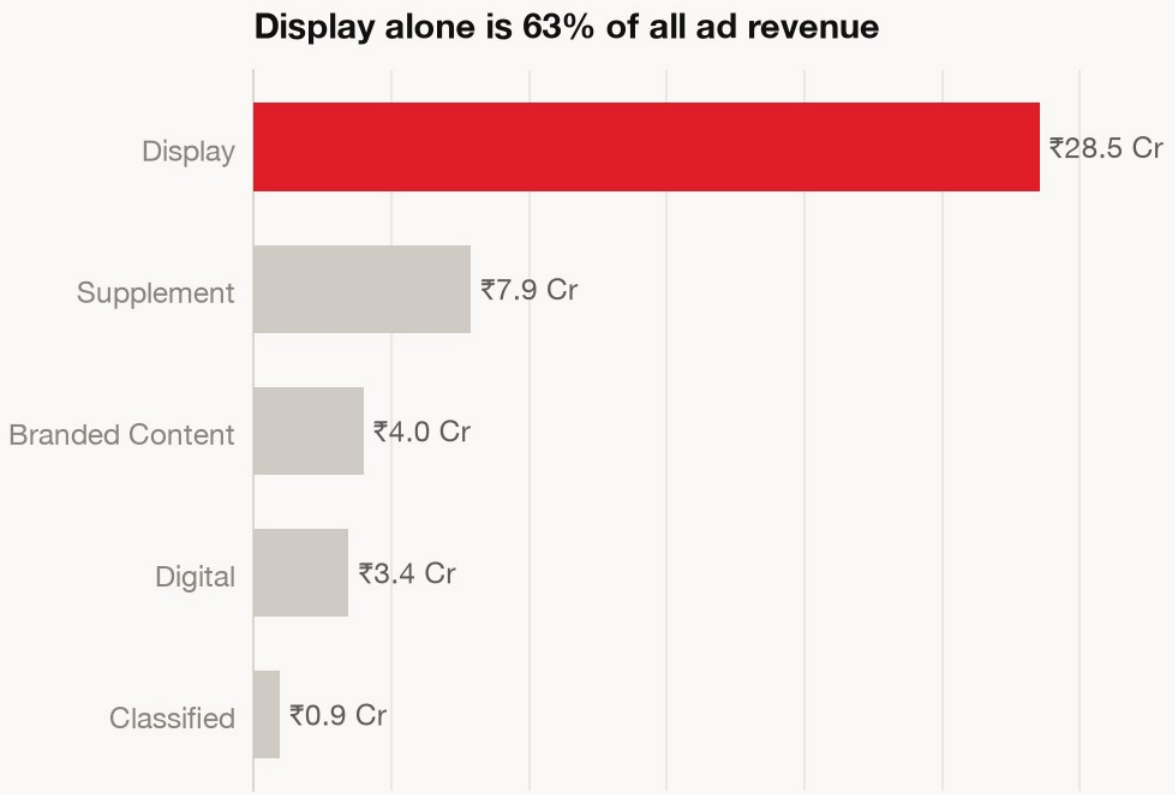
What are you trying to show?

"How things compare"	→	**BAR — horizontal, sorted**
"Who's best and worst"	→	**BAR — sorted, ends highlighted**
"How it moved over time"	→	**LINE**
"What makes up the total"	→	**STACKED BAR (pie only if ≤ 4 parts)**
"How values are spread"	→	**HISTOGRAM**
"Whether X and Y relate"	→	**SCATTER**
"What drove the change"	→	**WATERFALL**
"Patterns across two things"	→	**HEATMAP**
"Exact numbers, few rows"	→	**TABLE — not a chart**

Stuck? Use a horizontal bar chart. It is never the wrong answer.



A topic for a title, five colours, a legend to decode, no sort order



The finding as the title, one colour, sorted, labels on the bars

Three rules that fix almost every chart

Then point at anything still on the chart and ask what it proves. If it does not support the title, take it off.

1. **Put the finding in the title**
   “Display is 63% of revenue”, never “Revenue by Ad Type”.

2. **One colour carries the point**
   Everything else goes grey. Colour is an argument, not decoration.

3. **Label the bars, delete the furniture**
   Legend, gridlines, border, markers. All of it competes with the data.

MODULE 4 OF 5

04

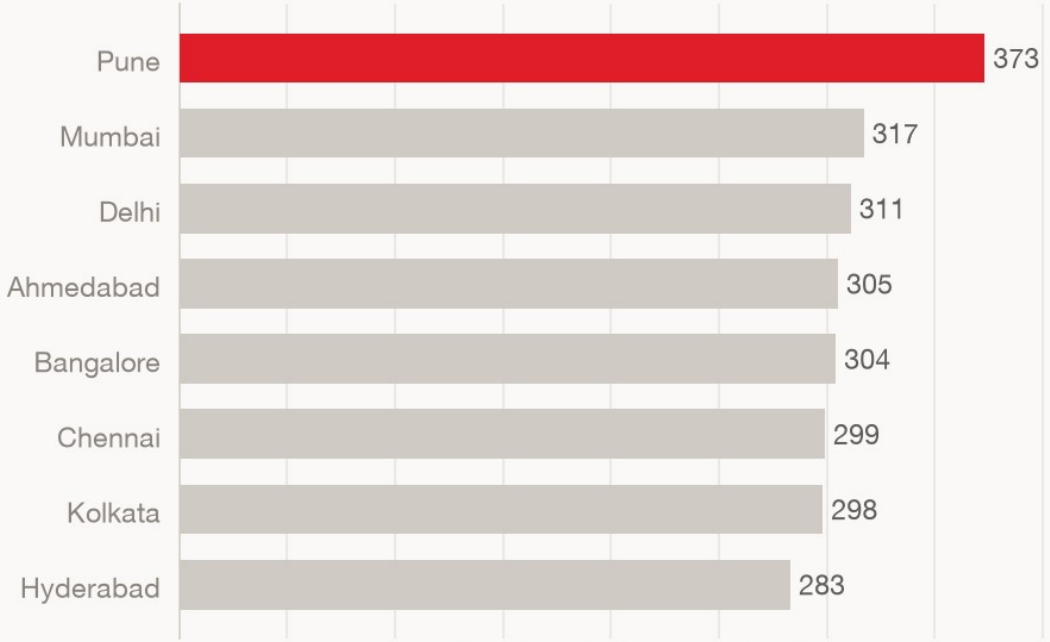
The Six Distortions

No formulas. Six ways a technically correct number ends up telling a room the wrong thing.

20 MINUTES

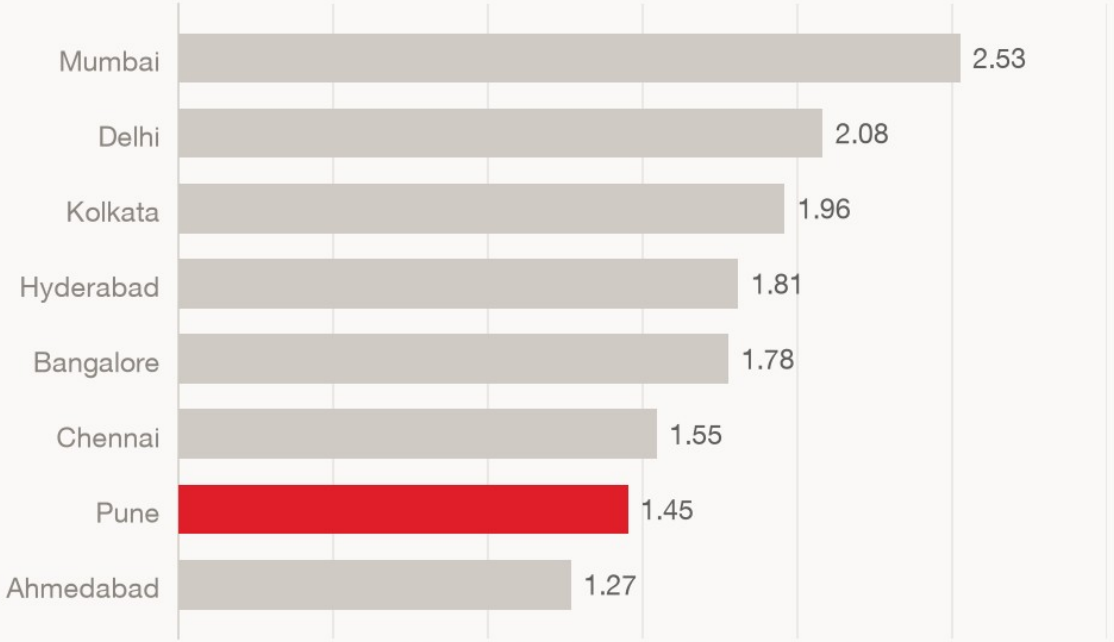
Pune tops one list and sits second-from-bottom on the other

COUNT: number of orders



Pune books the most orders

RATE: revenue per order (₹ lakh)



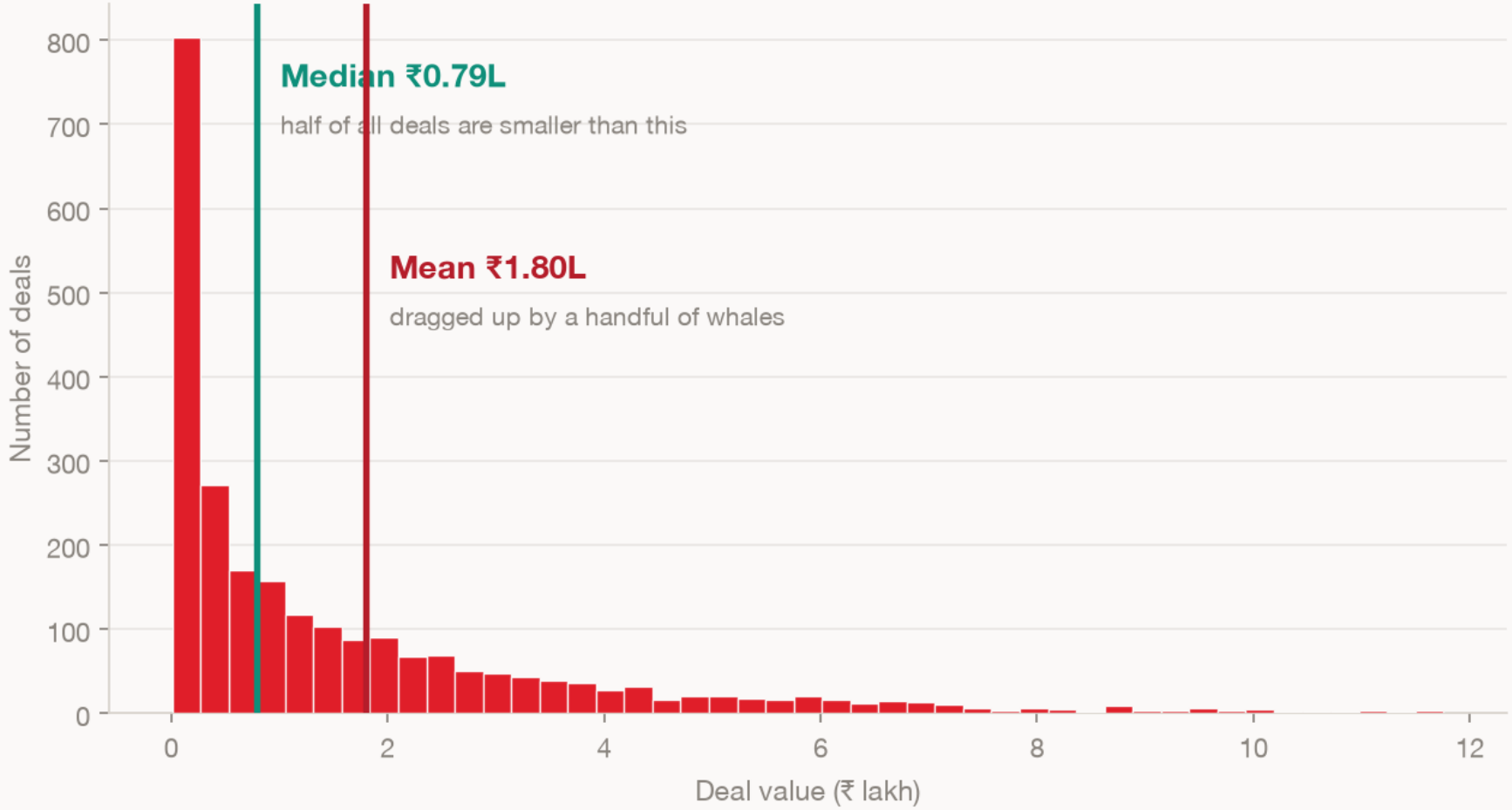
Pune is 7th of 8 on value per order

Always ask “per what?”

Counts reward whoever is biggest. Divide by something and you find out who is actually good at it. Per order, per rep, per rupee spent, per head.

The average deal is 2.3× the size of the typical deal

Confirmed deal value · ₹ lakh · tail beyond ₹12L trimmed so the shape is visible

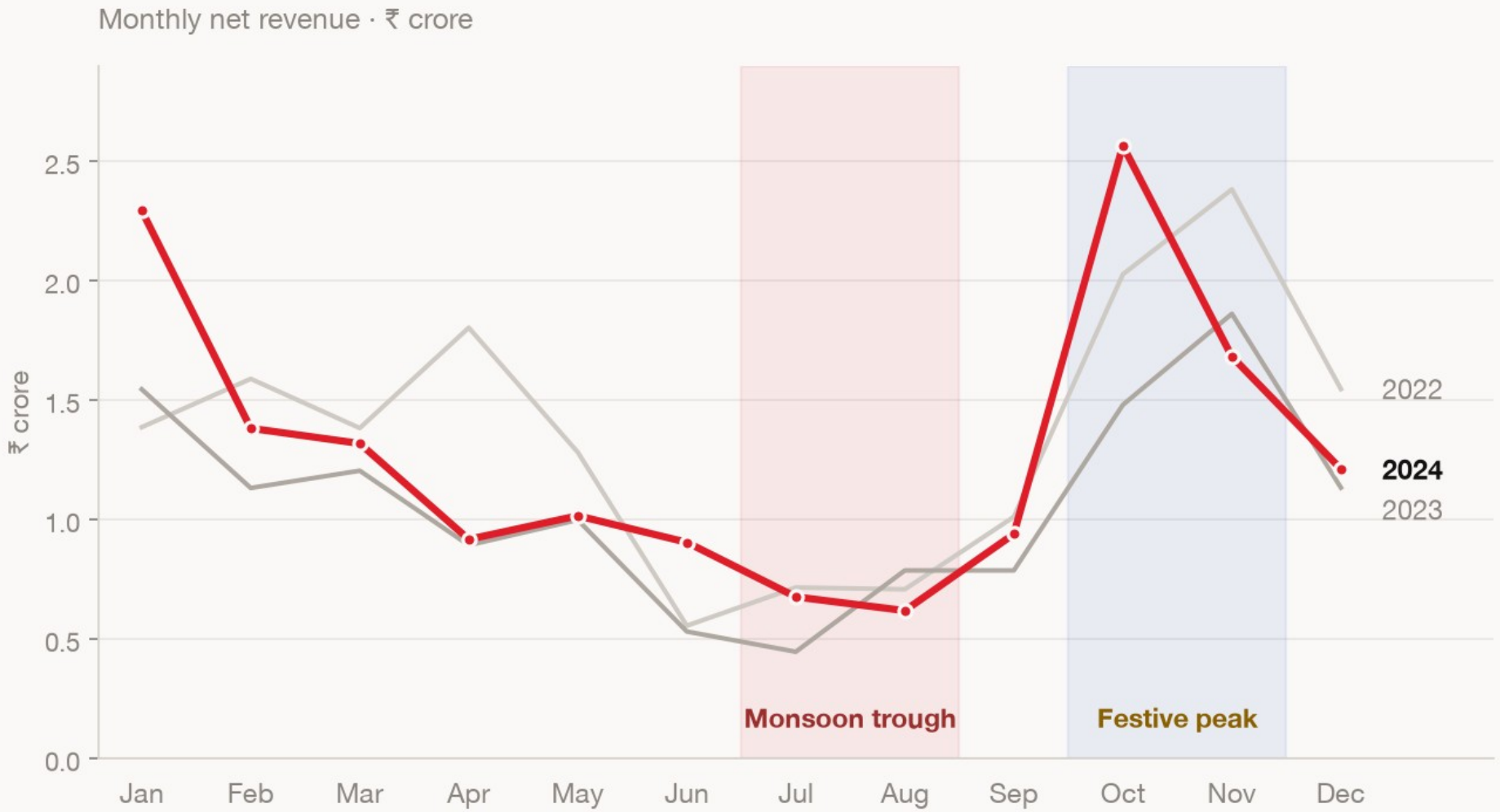


Mean, median, and the gap between them

Read the gap before you quote either

	WHAT IT IS	WHEN TO USE IT
Mean	Add everything up, divide by the count	Values cluster together with no long tail
Median	Line them up, take the one in the middle	A few large values are pulling the average
The gap	₹1.80L mean against ₹0.79L median	At 2.3× apart, quoting the mean overstates

October runs three times July, all three years



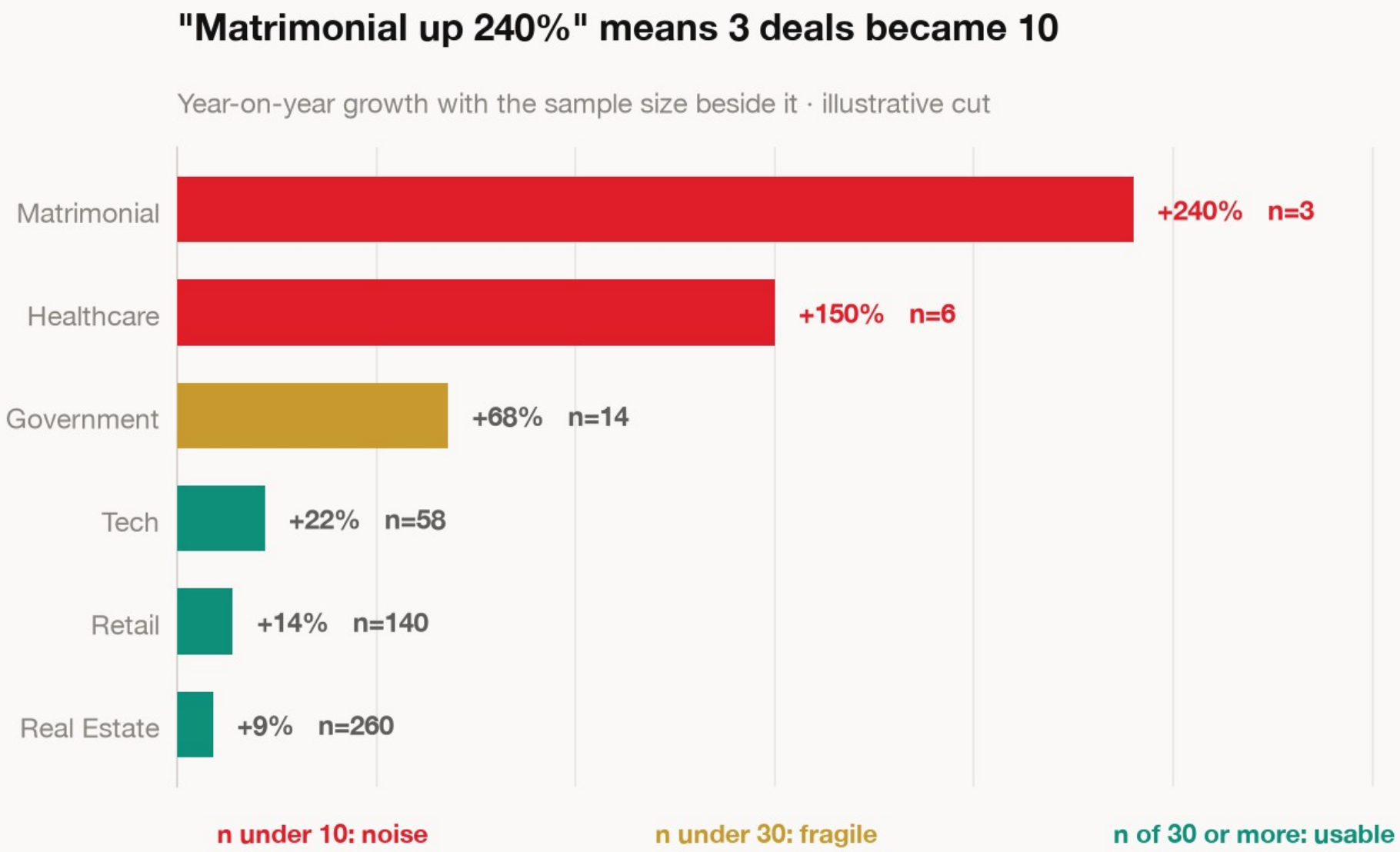
Which growth comparison to use

This file swings threefold between July and October, so a month-on-month number here is mostly calendar.

1. **Year on year**
   This January against last January. Cancels seasonality. Your default.

2. **Month on month**
   Fast signal, very noisy. Never multiply it by 12 to get an annual rate.

3. **A part-finished month**
   Exclude it, or put the day count right next to the number.


"Matrimonial up 240%" means 3 deals became 10
Year-on-year growth with the sample size beside it · illustrative cut
Matrimonial
+240% n=3
Healthcare
+150% n=6
Government
+68% n=14
Tech
+22% n=58
Retail
+14% n=140
Real Estate
+9% n=260
n under 10: noise
n under 30: fragile
n of 30 or more: usable

Distortion 5 — correlation reported as cause

Two series moving together is the beginning of the work, not the end

Ice cream sales and drowning deaths rise together.

Summer drives both.

SO WHEN YOU FIND A RELATIONSHIP

Ask what else changed in the same window

Campaigns, pricing, seasonality, a tracking change.

Say out loud whether you observed it or inferred it

Two words. It changes how the room treats the claim.

Go looking for the third thing driving both

That is usually where the actual explanation is.

WAR STORY · HT DIGITAL STREAMS · PLATFORM CONSOLIDATION

Three tools, three versions of the same metric.

Mixpanel, Redshift and Elasticsearch each held a version of the truth, and teams quoted whichever one they had access to. Meetings were spent reconciling numbers rather than deciding anything.

Consolidating onto one modelled layer with agreed definitions took most of six months. The reporting improvement was almost a side effect.

₹2.5 Cr

a year in run-rate savings once the duplicate stack was retired

Distortion 6 is the undefined metric. It is the cheapest one to fix and the one nobody owns.

MODULE 5 OF 5

05

The Five-Check Gate

Aurex will run your code perfectly and still answer the wrong question. This is the gate before you send.

15 MINUTES

A groupby on the wrong column runs perfectly

Nothing in the output tells you the column was wrong

You get a clean table, a tidy chart and a confident summary. The failure mode moved from arithmetic to aim.

MEASURED ACCURACY ON REAL SPREADSHEETS

Looking things up	89%
Filtering rows	71%
Adding things up	54%
Counting	42%
Sorting	38%
Medians	20%

Lookups come back right. Anything that needs arithmetic across rows, check it yourself.

The Five-Check Gate

94% of working spreadsheets already carry a human error. The bar here is checkability, not perfection.

1. **Row accounting**
   Ask how many rows it used, then compare that to how many are in the file.

2. **Spot-verify the headline**
   Pick the number that matters most and check that one against the source.

3. **Ask it a second way**
   Different phrasing, same question. Two answers means you have none.

4. **Open what it made you**
   Look at the axis. Read the labels. Generated files look finished early.

5. **Sign it**
   Would you put your name on this? You are about to.

HANDS-ON · 10 MIN

- Take the finding you produced earlier.
- Run all five checks against it.
- Rewrite it as one sentence your manager could act on.

Post in chat: “Your one sentence”

The five cards you are leaving with

Each one works on any file, in any tool, in any department

01	**THE SEVEN-STEP READ**	Context · Shape · Total · Break down · Compare · Drivers · Validate
02	**THE EIGHT LENSES**	Comparison · Ranking · Trend · Composition · Distribution · Correlation · Outliers · Decomposition
03	**THE CHART SELECTOR**	Name the comparison, take the chart it gives you
04	**THE SIX DISTORTIONS**	Rates · Mean · Calendar · Sample size · Cause · Definitions
05	**THE FIVE-CHECK GATE**	Rows · Headline · Second phrasing · Open the file · Sign it

Somebody sends you a spreadsheet you did not build,

about a part of the business you half-know,

and wants your read on it by tomorrow morning.

You now have a sequence for that.

Dishant Singla · Base42 Systems · run on Aurex, inside BCCL